

R Programming Tutorial

#07

Chi-Square (χ^2)

#lectureFIVE-TWO

Prepared by: Chan Weng Howe

#07-1

CHI-SQUARE TEST

ONE-WAY CONTINGENCY TABLE: EQUAL PROBABILITIES

One-way Contingency Table: Equal Probabilities

REFER to:

lectureSIX-TWO #02-1

EXAMPLE 1 (slide 21-27)

Example 1

A study was conducted on 147 cases of industrial accidents that required medical attention. Test the claim that the accidents occur with equal proportions on the 5 workdays.

Day	Mon	Tues	Wed	Thurs	Fri
Observed accidents	31	42	18	25	31

Chi-Square Test

One-way Contingency Table: Equal Probabilities

WHAT WE KNOW?

- One variable: No. of Accidents
- Categories: 5 (Mon-Fri)
- Significance level: 0.05

WHAT WE WANT TO FIND?

$H_0: P_1=P_2=P_3=P_4=P_5$

H_1 : At least 1 of the 5 proportions is different from others

Function in R:

`chisq.test(observed_frequency,
correct=FALSE)`

In R, `chisq.test()` by default applies "Yates" correction on the chi-square test. Which is arguably unnecessary, so have to set `correct=FALSE`

observed data

```
1 noAcc <- c(31, 42, 18, 25, 31)
2
3 output <- chisq.test(noAcc,
4
5 correct=FALSE)
```

Chi-squared test for given probabilities

data: noAcc
X-squared = 10.653, df = 4, p-value = 0.03075

X² statistics of the
observed data

P-value of the X²
statistics

Chi-Square Test

One-way Contingency Table: Equal Probabilities

Function in R:

```
chisq.test(observed_frequency,
           correction=FALSE)
```

Save result into

output

```
> str(output)
List of 9
 $ statistic: Named num 10.7
 .. attr(*, "names")= chr "X-squared"
 $ parameter: Named num 4
 .. attr(*, "names")= chr "df"
 $ p.value: num 0.0308
 $ method: chr "Chi-squared test for given probabilities"
 $ data.name: chr "noAcc"
 $ observed: num [1:5] 31 42 18 25 31
 $ expected: num [1:5] 29.4 29.4 29.4 29.4 29.4
 $ residuals: num [1:5] 0.295 2.324 -2.102 -0.811 0.295
 $ stdres: num [1:5] 0.33 2.598 -2.351 -0.907 0.33
 - attr(*, "class")= chr "htest"
```

str() function use to inspect the structure of a variable in R

```
> output$statistic
X-squared
10.65306

> output$p.value
[1] 0.03075267

> output$parameter
df
4

> output$observed
[1] 31 42 18 25 31

> output$expected
[1] 29.4 29.4 29.4 29.4 29.4
```

Variable in R could be a "List", to access the specific information from the list can use the \$ sign follow by the information name

observed data

```
1 noAcc <- c(31, 42, 18, 25, 31)
2
3 output <- chisq.test(noAcc,
4
5 correct=FALSE)
```

Chi-squared test for given probabilities

data: noAcc
X-squared = 10.653, df = 4, p-value = 0.03075

X² statistics of the
observed data

P-value of the X²
statistics

Chi-Square Test

One-way Contingency Table: Equal Probabilities

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

WHAT WE KNOW?

- One variable: No. of Accidents
- Categories: 5 (Mon-Fri)
- Significance level: 0.05

WHAT WE WANT TO FIND?

$H_0: P_1=P_2=P_3=P_4=P_5$

H_1 : At least 1 of the 5 proportions is different from others

Function in R:

qchisq(alpha, df, lower.tail=FALSE)

- To get the x2 statistics for the corresponding alpha values/p-values.

pchisq(x2, df, lower.tail=FALSE)

- To get the p-value for the corresponding x2 statistics values.

Goodness-of-Fit test in chi-square always right-tail(upper-tail), so need to set the parameter "lower.tail=FALSE"

```

1  noAcc <- c(31, 42, 18, 25, 31)
2  expprob <- sum(noAcc)/5
3  expAcc <- c(expprob, expprob,
4  expprob, expprob, expprob)
5
6  #calculate x2 for each category
7  exp <- ((noAcc-expAcc)^2)/expAcc
8
9  #sum up to a x2 statistics
10     x2 <- sum(exp)
11
12     #critical value
13     alpha <- 0.05
14     x2.alpha <- qchisq(alpha,
15     df=4,
16     lower.tail=FALSE)
17
18     #value of x2 of data
19     x2.data <- sum((noAcc-expprob)^2/expprob)
20
21     #p-value
22     p.value <- pchisq(x2.data,
23     df=4,
24     lower.tail=FALSE)
  
```

#07-2

CHI-SQUARE TEST

ONE-WAY CONTINGENCY TABLE: UNEQUAL PROBABILITIES

One-way Contingency Table: Unequal Probabilities

REFER to:

lectureSIX-TWO #02-2

EXAMPLE 4 (slide 40-44)

Example 4

Mars, Inc. claims its M&M candies are distributed with the color percentages of 30% brown, 20% yellow, 20% red, 10% orange, 10% green, and 10% blue. At the 0.05 significance level, test the claim that the color distribution is as claimed by Mars, Inc. The observed frequency as shown below:

Frequencies of M&Ms candies

	Brown	Yellow	Red	Orange	Green	Blue
Observed frequency	33	26	21	8	7	5

Chi-Square Test

One-way Contingency Table: Unequal Probabilities

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

WHAT WE KNOW?

- One variable: Frequencies of M&Ms
- Categories: 6 colors
- Significance level: 0.05
- Claimed p:

$p_1 = 0.30, p_2 = 0.20, p_3 = 0.20, p_4 = 0.10, p_5 = 0.10, p_6 = 0.10$

WHAT WE WANT TO FIND?

$H_0: p_1 = 0.30, p_2 = 0.20, p_3 = 0.20, p_4 = 0.10, p_5 = 0.10, p_6 = 0.10$

H_1 : At least 1 proportions is different from claimed

Function in R:

`chisq.test(observation, p=prob, correct=FALSE)`

In R, `chisq.test()` by default applies "Yates" correction on the chi-square test. Which is arguably unnecessary, so have to set **correct=FALSE**

Since the claim has unequal probabilities, set the probability into the parameter **p** of the function

```
1  mnm <- c(33, 26, 21, 8, 7, 5)
2  prob <- c(0.3, 0.2, 0.2, 0.1,
3      0.1, 0.1)
4
5  # perform chi-square test on the
6  data with their correspond
7  probabilities
8  chisq.test(mnm, p=prob,
9  correct=FALSE)
10
11  #critical value > x2.alpha
12  alpha <- 0.05 [1] 11.0705
13  x2.alpha <- qchisq(alpha,
14  df=5,
15  p=0.975)
16
17  > chisq.test(mnm, p=prob, correct=FALSE)

Chi-squared test for given probabilities
```

```
data: mnm
X-squared = 5.95, df = 5, p-value = 0.3111
```

#07-3

CHI-SQUARE TEST

TWO-WAY CONTINGENCY TABLE: TEST OF INDEPENDENCE

REFER to:

lectureSIX-TWO #03

EXERCISE 3 (slide 69)

EXERCISE #3

Jail inmates can be classified into one of the following four categories according to the type of crime committed: violent crime, crime against property, drug offenses, and public-order offenses. Suppose that random samples of 500 male inmates and 500 female inmates are selected, and each inmate is classified according to type of offense.

Type of Crime	Gender	
	Male	Female
Violent	117	66
Property	150	160
Drug	109	168
Public-order	124	106

We would like to know whether male and female inmates differ with respect to type of offense. Test the relevant hypotheses using a significance level of 0.05.

Chi-Square Test

Two-way Contingency Table: Test of Independence

WHAT WE KNOW?

- Two variables:
 - Gender: Male / Female
 - Type of Crime: 4 categories
- Significance level: 0.05

WHAT WE WANT TO FIND?

H_0 : Crime type is independent of gender

H_1 : Crime type is not independent of gender

```
1  male <- c(117,150,109,124)
2  female <- c(66,160,168,106)
3  d <- data.frame(male, female)
4
5  # perform chi-square test on the
6  data table
7  chisq.test(d, correct=FALSE)
8
9  #critical value > x2.alpha
10 alpha <- 0.05 [1] 7.814728
11 x2.alpha <- qchisq(alpha,
12 df=3,
13 lower.tail=FALSE)
14
15 > chisq.test(d,correct=FALSE)
16
17      Pearson's Chi-squared test
18
19 data:  d
20 X-squared = 28.511, df = 3, p-value = 2.837e-06
21
22
```

Two-way Contingency Table: Test of Independence

EXAMPLE

In the built-in data set survey, the **Smoke** column records the students smoking habit, while the **Exer** column records their exercise level.

The allowed values in Smoke are "Heavy", "Regul" (regularly), "Occas" (occasionally) and "Never". As for Exer, they are "Freq" (frequently), "Some" and "None".

Test the hypothesis whether the students smoking habit is independent of their exercise level at .05 significance level.

Chi-Square Test

Two-way Contingency Table: Test of Independence

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

Function in R:

`table(list1, list2,...)`

- Group lists into a contingency table

```

1  library(MASS)
2  # get the contingency table
3  tbl = table(survey$Smoke,
4             survey$Exer)
5
6
7  # perform chi-square test on the
8  data table
9  chisq.test(tbl, correct=FALSE)
10
11     #critical val > x2.alpha
12     alpha <- 0.05 [1] 7.814728
13     x2.alpha <- qchisq(alpha,
14                       df=3,
15                       lower.tail=FALSE)
16
17     > chisq.test(tbl,correct=FALSE)
18
19     Pearson's Chi-squared test
20
21     data:  tbl
22     X-squared = 5.4885, df = 6, p-value = 0.4828
23
24     Warning message:
25     In chisq.test(tbl, correct = FALSE) :
26       Chi-squared approximation may be incorrect
  
```

Chi-Square Test

Two-way Contingency Table: Test of Independence

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

```
> tbl
```

	Freq	None	Some
Heavy	7	1	3
Never	87	18	84
Occas	12	3	4
Regul	9	1	7

Warning from the chisquare could be due to the small cell values in the contingency table

```

1  library(MASS)
2  # get the contingency table
3  tbl = table(survey$Smoke,
4             survey$Exer)
5
6
7
8  # perform chi-square test on the
9  data table
10 chisq.test(tbl, correct=FALSE)
11
12 #critical val > x2.alpha
13 alpha <- 0.05 [1] 7.814728
14 x2.alpha <- qchisq(alpha,
15                    df=3,
16                    lower.tail=FALSE)
17
18 > chisq.test(tbl,correct=FALSE)
19
20 Pearson's Chi-squared test
21
22 data:  tbl
23 X-squared = 5.4885, df = 6, p-value = 0.4828
24
25 Warning message:
26 In chisq.test(tbl, correct = FALSE) :
27   Chi-squared approximation may be incorrect
  
```

Chi-Square Test

Two-way Contingency Table: Test of Independence

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

```
> tbl2 = cbind(tbl[, 'Freq'], tbl[, 'None'] + tbl[, 'Some'])
> tbl2
```

	[,1]	[,2]
Heavy	7	4
Never	87	102
Occas	12	7
Regul	9	8

Create another table, by
merging the 2nd and 3rd columns
into single columns

```
1 library(MASS)
2 # get the contingency table
3 tbl = table(survey$Smoke,
4 survey$Exer)
5 # create a new table
6 tbl2 = cbind(tbl[, 'Freq'],
7 tbl[, 'None'] + tbl[, 'Some'])
8
9 # perform chi-square test on the
10 data table
11 chisq.test(tbl2, correct=FALSE)
12
13 #critical val> x2.alpha
14 alpha <- 0.05[1] 7.814728
15 x2.alpha <- qchisq(alpha,
16 df=3,
17 lower.tail=FALSE)
```

```
> chisq.test(tbl2, correct=FALSE)
```

```
Pearson's Chi-squared test
```

```
data: tbl2
```

```
X-squared = 3.2328, df = 3, p-value = 0.3571
```

THE END...